



4.4 Probability, Venn and tree diagrams

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Probabilities satisfy $0 \leq P(A) \leq 1$, with $P(A') = 1 - P(A)$. For combined events, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (reducing to $P(A) + P(B)$ if mutually exclusive). Events A and B are independent if $P(A \cap B) = P(A)P(B)$; conditional probability is $P(A|B) = P(A \cap B)/P(B)$. Venn diagrams organize overlapping events, while tree diagrams model sequences of events (e.g. sampling with or without replacement), multiplying along branches and adding across separate paths.

Worked example: follow the reasoning

In a class of 30 students, 18 study French, 14 study Spanish, and 8 study both. Find the probability a randomly selected student studies (a) French or Spanish, (b) neither.
Solution: (a) $P(F \cup S) = 18/30 + 14/30 - 8/30 = 24/30 = 0.8$. (b) $P(\text{neither}) = 1 - 0.8 = 0.2$.

Quick check · multiple choice

If $P(A) = 0.4$, $P(B) = 0.5$, and A, B are independent, find $P(A \cap B)$.

- A 0.9
- B 0.2
- C 0.1
- D 0.7

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A bag contains 5 red and 3 blue balls. Two balls are drawn at random without replacement. (a) State the probabilities on the first set of branches of a tree diagram for this situation, and the four second-draw conditional probabilities. (b) Find the probability both balls are red. (c) Find the probability that at least one ball is blue. (d) Given that the second ball drawn is red, find the probability that the first ball drawn was also red. [8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) First draw: $P(R)=5/8$, $P(B)=3/8$. If first is red, remaining is 4R,3B: $P(R|R)=4/7$, $P(B|R)=3/7$. If first is blue, remaining is 5R,2B: $P(R|B)=5/7$, $P(B|B)=2/7$. (b) $P(RR)=5/8 \times 4/7 = 5/14$. (c) $P(\text{at least one blue}) = 1 - 5/14 = 9/14$. (d) $P(2^{\text{nd}} R) = P(RR) + P(BR) = 5/14 + (3/8 \times 5/7) = 20/56 + 15/56 = 35/56 = 5/8$. So $P(1^{\text{st}} R \mid 2^{\text{nd}} R) = (5/14)/(5/8) = 4/7$.

Quick check answer

B. For independent events, $P(A \cap B) = P(A) \times P(B) = 0.4 \times 0.5 = 0.2$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.